POSTER PRESENTATION

# Forecasting USD/INR in the Indian Market with VMD, GDELT Agents, Monte Carlo, and TEPC

Repository summary of how the project moved from raw news signals toward bounded LLM overlays, market-memory systems, and topology-enabled predictions from chaos.


COURSE IN CHARGE
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POSTER FORMAT
**A4 Portrait**

- Problem in Indian FX Market
- VMD + Statistical Signal Recovery
- Monte Carlo Randomness Check
- GDELT Multi-Agent LLMs
- TEPC on Chaotic Data
- TEPC + LLM Hybrid Results

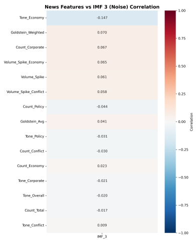
**RAW MARKET SIGNAL**
**0.010**
Goldstein vs IMF3 is near zero.

**RECOVERED SIGNAL**
**-0.147**
Theme-filtered Tone_Economy recovers signal.

**V4 LLM RESULT**
**64.17%**
Gemini V4 direction accuracy on 240 days.

**TEPC RESULT**
**77.5%**
Strict TEPC breakout accuracy. Macro-F1 = 0.337.

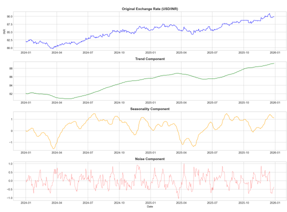
## 1. PROBLEM
### Why USD/INR Is Hard to Predict

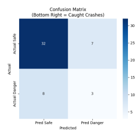


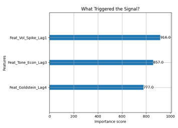

- USD/INR is shaped by RBI intervention, oil shocks, and dollar cycles.
- Raw sentiment is weak: Goldstein vs IMF3 = **0.010**; US tone vs USD/INR = **0.164**.
- Delay still matters: Goldstein peaks at **9 days**; tone peaks at **12 days**.

Key insight: raw aggregates look random, but filtered themes recover structure.

## 2. VMD + STATS
### VMD Turns Price into Volatility

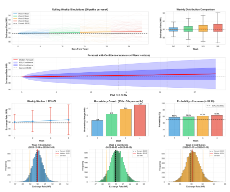


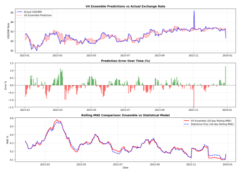

- VMD targets **IMF3**, the high-frequency part of USD/INR.
- Phase B uses **Economy**, **Conflict**, **Policy**, and **Corporate** themes.
- **Tone_Economy = -0.147**; Phase-C reaches **40% precision** and **52% recall**.

## 3. MONTE CARLO
### Monte Carlo as a Randomness Check



- Monte Carlo works best as a **risk engine**, not a point predictor.
- Best 30-day setup: regime switching with median **90.370** and std **1.336**.

## 4. GDELT AGENTS
### Bounded GDELT LLM Overlay



**V4 ACCURACY TABLE**

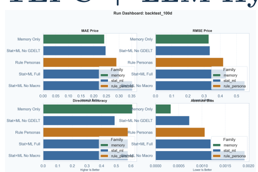
| Model | Days | MAE | Dir. |
|---|---|---|---|
| V3 hybrid | 240 | 0.2741% | 61.25% |
| V4 hybrid | 240 | **0.2695%** | **64.17%** |
| V4 Jan live | 21 | 0.3216% | 57.14% |

Debiasing improved V4 through symmetric prompts, random order, and adaptive weights.

## 5. TEPC
### TEPC: Topology and Chaos Results


- TEPC models USD/INR inside a macro network with DXY, BRENT, GOLD, rates, and sentiment.
- Graph features and coupled Lorenz chaos features become predictors.
- **tepc_full**: **77.5%** breakout accuracy, **0.337** macro-F1, **0.002746** MAE return. Live **chaos_only** reaches **80%**.

## 6. HYBRID TAKEAWAY
### TEPC + LLM Hybrid Results



**HYBRID TABLE**

| Model | Window | Acc. | Error |
|---|---|---|---|
| Memory-only | 100d | 61.0% dir. | 0.2421 price |
| Gemini 2.5 | 100d | 58.0% dir. | 0.2726 price |
| TEPC full | 80d | 77.5% brk. | 0.002746 ret. |
| TEPC blend | 80d | **78.75% brk.** | **0.002695 ret.** |

LLMs help most with structured market memory, and TEPC adds regime signal on top.

**Final Message:** best results come from **VMD + GDELT filtering + bounded LLM overlays + TEPC topology/chaos features**, not from a larger black-box model.